Explaining Rates Using Collision Theory

Exam code: 0620 & 0971

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Explaining rates using collision theory

Extended tier only

- Collision theory can explain why concentration, pressure, surface area, temperature and catalysts affect the rate of a chemical reaction

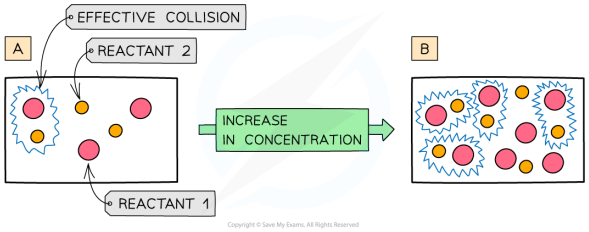
How increasing concentration affects rate

- Increasing the concentration of a solution increases the rate of reaction
- Increasing the concentration means that there are more reactant particles in a given volume
  - This causes more collisions per second
  - Leading to more frequent and successful collisions per second
  - Therefore, the rate of reaction increases
- If you double the number of particles, you will double the number of collisions per second
  - The number of collisions is **proportional** to the number of particles present

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



A higher concentration of particles in (b) means that there are more particles present in the same volume than (a) so the number of collisions and successful collisions between particles increases causing an increased rate of reaction

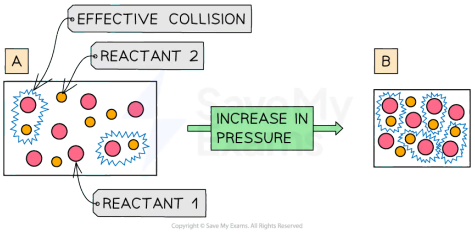
Examiner Tips and Tricks

When answering questions on the effect of concentration on the rate of reaction, you should mention that there are more particles **per unit volume** (usually cm^3) and this causes an increase in the rate of collisions.

How increasing pressure affects rate

- Increasing the pressure of a gas increases the rate of reaction
- Increasing the pressure means that there are the same number of reactant particles in a smaller volume
  - This causes more collisions per second
  - Leading to more frequent and successful collisions per second
  - Therefore, the rate of reaction increases

Diagram showing the effect of increasing pressure

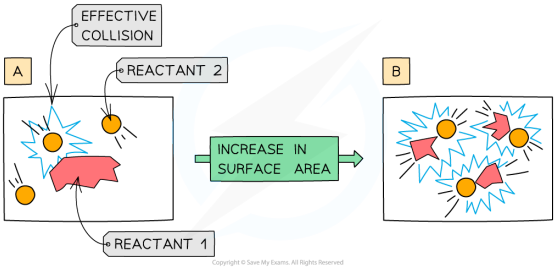


The higher pressure (b) means that there are the same number of particles present in a smaller volume than (a) so the number of collisions and successful collisions between particles increases causing an increased rate of reaction

- Increasing the surface area increases the rate of reaction
- Increasing the surface area means that a greater **surface area** of **particles** will be exposed to the other reactant
  - This causes more collisions per second
  - Leading to more frequent and successful collisions per second
  - Therefore, the rate of reaction increases

- If you double the surface area, you will double the number of collisions per second

Diagram showing the effect of increasing surface area



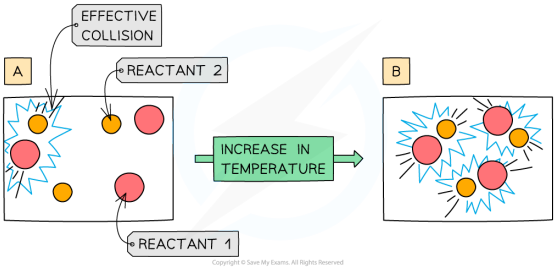
An increase in surface area means more collisions per second

How increasing temperature affects rate

- Increasing the temperature increases the rate of reaction
- Increasing the temperature means that the particles have more kinetic energy
  - This causes more collisions per second
  - Leading to more frequent and successful collisions per second
  - Therefore, the rate of reaction increases

- The effect of temperature on collisions is not so straightforward as concentration or surface area; a small increase in temperature causes a large increase in rate
- For aqueous and gaseous systems, a rough rule of thumb is that for every 10 °C increase in temperature, the rate of reaction approximately doubles

Diagram showing the effect of increasing temperature



How a catalyst affects rate

- Catalysts are substances which **speed up** the **rate** of a reaction without themselves being **altered** or **consumed** in the reaction
- Different processes require different types of catalysts but they all work on the same principle
- A catalyst provides a **different reaction pathway** that has a **lower activation energy**
  - This means a higher proportion of the reactant particles have energy greater than the activation energy and will result in more successful collisions per second

A catalyst lowers the activation energy of a reaction by providing an alternative reaction pathway

- An important industrial example is iron, which is used to catalyse the Haber Process for the production of ammonia
  - Iron **beads** are used to increase the **surface area** available for catalysis
- Enzymes are **biological** catalysts, they work best at specific temperature and pH ranges
- Normally only **small amounts** of catalysts are needed to have an effect on a reaction

Examiner Tips and Tricks

- Temperature is the only factor that **directly** affects the energy of collisions because the increased thermal energy is converted to increased kinetic energy in the particles
  - Concentration, pressure and surface area only affect the number of collisions
- Temperature affects reaction rate by increasing the number of collisions and increasing the energy of the collisions
  - Of these two, the increase in energy is the more important one.

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